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MA-1841

B. Tech. (First Semester)

EXAMINATION, 2025-26

MATHEMATICS-I

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Define commutative property of partial differential coefficients of function $f(x, y)$.

P. T. O.

~~2.~~ Expand $\sin x$ by Maclaurin's theorem. ✓

~~3.~~ Discuss the Jacobian of u, v with respect to x, y .

~~4.~~ If A is a real skew matrix such that $A^2 + I = O$, show that A is orthogonal. ✓

✓ 5. Write the condition for consistency of system of equations $AX = B$.

~~6.~~ State Cayley-Hamilton theorem. ✓

7. Prove that :

$$\sqrt{\frac{1}{2}} = \sqrt{\pi}. \quad \checkmark$$

~~8.~~ Explain the term quadrature.

~~9.~~ Show that :

$$\nabla(fg) = f(\nabla g) + g(\nabla f).$$

✓ 10. Define the surface and volume integrals.

(5-6)

$$\boxed{AX = B}.$$

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

✓ If:

$$\mu = \log \frac{x^4 + y^4}{x + y}$$

show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$.

✓ Find the rank of matrix :

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Euler's

3. Evaluate $\iint (x^2 + y^2) dx dy$ over the region in the positive quadrant for which $x + y \leq 1$.

Euler's

✓ 4. Prove that :

$$\text{div grad } r^m = m(m+1)r^{m-2}$$

Euler's

P. T. O.

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Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

✓ 1. If $y = e^{a \sin^{-1} x}$, prove that :

$$(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} -$$

$$(n^2 + a^2)y_n = 0$$

and hence find the value of y_n at $x = 0$.

2. Find the dimensions of the rectangular parallelepiped of maximum volume that can be cut from the ellipsoid :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

3. Investigate for what values of λ, μ the simultaneous equations :

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

have (i) no solution (ii) a unique solution.

4. Verify Green's theorem in the plane for

$$\int_C \left\{ (3x^2 - 8y^2) dx + (4y - 6xy) dy \right\} \text{ where}$$

C is the region bounded by the parabola

$$y = \sqrt{x} \text{ and } y = x^2.$$

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